5.1.4 Infrastructure and Economic Growth

Infrastructural facilities are vital for the smooth functioning of the economy. They are like wheels of development without which the economy will not be able to function properly. Provision of adequate infrastructure, both in terms of quantity and quality, is essential for sustenance of economic growth. They contribute to improvement in factor productivity and by providing amenities that enhance the quality of life. The role of infrastructure development in economic growth has been well recognised in the literature. Discussed below are some of the most critical significance of economic infrastructure and its impact on the economy.

1) *Productivity and Growth:* Better quantity and quality of infrastructure can directly raise the productivity of human and physical capital and hence growth (e.g. by providing access, roads can: (i) improve education and markets for farmers' outputs and others by cutting costs, (ii) facilitate private investment, (iii) improve jobs and income levels for many). Studies reveal that 1 per cent growth in the stock of infrastructure often associates with 1 per cent growth in per capita GDP.

2) *Contributes to Domestic Market Development:* Infrastructural development also contributes to domestic market development. Rural roads in the developing countries have a major effect in improving marketing opportunities and reducing transaction costs. The marketing of agricultural commodities can account for 25-60 per cent of final prices of food stuffs in developing countries. About half of the marketing costs are attributable to transport. Thus, proper infrastructure can reduce the cost of marketing which in turn contributes to domestic market development.

3) *Infrastructure mobilises Savings (Domestic and Foreign).* Infrastructure development typically involves high levels of upfront financial investment, given the heavy capital expenditure required during the construction stage. Thus, mobilising savings for infrastructure development becomes indispensable. In the globalised era infrastructure facilities plays an important role in attracting foreign capital. Foreign direct investment as well as portfolio investment flows to those countries where adequate infrastructure facilities are available.

4) *Infrastructure Generates Employment Opportunities:* Infrastructure is a base for larger investment, development of industry and agriculture which in turn creates more employment opportunities. Infrastructure improves mobility, productivity and efficiency of labour.

5) *Infrastructure and Social Development:* Infrastructure increases the employment opportunities which increase the income level of the people further enhancing the standard of living of the people. Thus, infrastructure development will change the total outlook of the society and will lead to social development.

6) *Assists to Reduce Poverty:* Creation of infrastructure helps to reduce poverty by improving the quality of life. Access to clean water and sanitation has direct consumption benefits in reducing mortality and morbidity that increases the productive capacity of the poor. Access to transport can

contribute to higher and more stable incomes enabling the poor to manage risk. Transport infrastructure has been found to expand opportunities for non-farm employment in rural areas. Improved rural transport can also ease the introduction of improved farming practices by lowering the costs of modern inputs.

7) *Promotes Social Change:* Infrastructural facilities also act as an instrument of social change. Development of industries, transport facilities and education changes the outlook of people. Apart from these, even science, technology and growth of towns and cities will lead to a changed economic outlook.

8) *Infrastructure and National Defence:* Infrastructure development in terms of proper transport and communication facilities, adequate facilities for research and development, proper military schools etc. increase the strength of the nation from defence perspective.

5.1.5 Role of Public Private Partnership in Infrastructure

Infrastructure may be owned and managed by governments or by private companies, such as a public utility or railway companies. Traditionally most roads, major airports and other ports, water distribution systems, and sewage networks have been publicly owned, but this is changing as now one finds many energy and telecommunications networks are privately owned, so are airports, roads and civic infrastructure. Government-owned and operated infrastructure may be developed and operated in the private sector or in public-private partnerships. Besides supplementing the public resources, public-private partnerships (PPPs) enable the public sector to benefit from private sector technical expertise, experience and efficiency, and transfer project-related risks to the private sector. In infrastructure, PPPs have helped addressing financial gaps. However, the actual performance of infrastructure sector PPPs has been mixed. While a few sectors like roads and highways and airports have been able to attract significant private investments, most others like ports, water treatment and recycling, health and education have seen limited traction. Over the last decade, the share of private investments in the sector fell from 37 per cent in 2008 to about 25 per cent in 2018. A number of important steps have been initiated by the government to provide a boost to infrastructure PPPs over last few years.

One of the most significant has been the setting up of the National Investment and Infrastructure Fund (NIIF), with the government contributing around Rs. 20,000 crore. A number of policy-related measures aimed at improving liquidity in infrastructure investments have also been taken such as announcement of Infrastructure Investment Trust guidelines by SEBI in September 2015, promulgation of the Insolvency and Bankruptcy Code, 2016 and the November, 2018 directive by SEBI to large corporates to fund at least 25 per cent of their borrowings through the corporate bond market with effect from Financial Year (FY) 19-20.

While these initiatives are steps in the right direction, there are a few other areas which need to be addressed on a priority basis given the global experience in public private partnerships. Firstly, there is a need to consolidate and update the existing PPP guidelines based on the experience gained across individual sectors over the last few years as well as best practices in other countries. Secondly, need to formulate an enabling framework to renegotiate concession agreements. Finally, there is an urgent need for a dispute-resolution process for

Public Private Partnership (PPP) arrangements, as long-pending disputes have significantly adverse financial impact and act as a deterrent for private partners.

5.1.6 Budget Allocations for Infrastructure Sector

The social and economic transformation of society depends on availability of inclusive and sustainable infrastructure amenities to the people and the ability of the economy to address its infrastructure bottlenecks. In the past decade (FY 2008-17), India invested about $1.1 trillion on infrastructure. The challenge is to step-up annual infrastructure investment so that lack of infrastructure does not become a binding constraint on the growth of the Indian economy. In order to reach the GDP target of $5 trillion by 2024-25, India needs to spend about $1.4 trillion (Rs. 100 lakh crore) over these years on infrastructure, including social and economic infrastructure projects. To achieve this objective, a Task Force was constituted to draw up the National Infrastructure Pipeline (NIP) for each of the years from FY 2019-20 to FY 2024-25. NIP is set to enable a forward outlook on infrastructure projects which will create jobs, improve ease of living, and provide equitable access to infrastructure for all, thereby making growth more inclusive. As per the detailed reports of the task force, total project capital expenditure in infrastructure sectors in India during the fiscals 2020 to 2025 is projected at over Rs. 102 lakh crore, around 70 per cent of this is amounted by sectors such as Energy (24%), Roads (19%), Urban (16%), and Railways (13%).

5.1.7 Social and Physical Infrastructure

A country's level of human and economic development is closely related to its levels of achievement in physical and social infrastructure. While physical infrastructure is an important determinant of domestic production, good social infrastructure is vital for human development as well as economic progress through better educated, better skilled, and healthier citizens, or citizens with better human capabilities.

Social infrastructure

Social infrastructure refer to basic services such as health, education, skill formation and training that contribute to human resource development and its quality. It also includes sanitation, drinking water, housing, sewerage, etc. Social infrastructures are also termed as social overheads. These social overheads indirectly support the economic systems as they contribute to productivity improvements. Besides meeting some social objectives, the social infrastructure supports growth in the long run. For example, consuming education services does not directly affect economic activities like production and distribution of goods, but indirectly helps in the economic development of the country by producing the skill and knowledge sets in the form of scientists, technologists and engineers, who contribute to economic activity. Social infrastructure enhances social well-being and furthers economic growth by providing basic services and facilities which allow businesses to develop and flourish.

Trends in Social Sector Expenditure

As per Economic Survey 2020-21, the expenditure on social services (education, health and other social sectors) by the Centre and States combined as a proportion of GDP increased from 6.2 to 8.8 per cent during the period 2014-15 to 2020-21 (BE). This increase was witnessed across all social sectors. For education, it increased from 2.8 per cent in 2014-15 to 3.5 per cent and for health, from 1.2 per cent to 1.5 per cent during the same period. Refer Table 5.1.